HR Analytics: Workforce & Attrition Insights

Power BI Project — IBM HR Analytics Dataset

PEOPLE ANALYTICS

POWER BI

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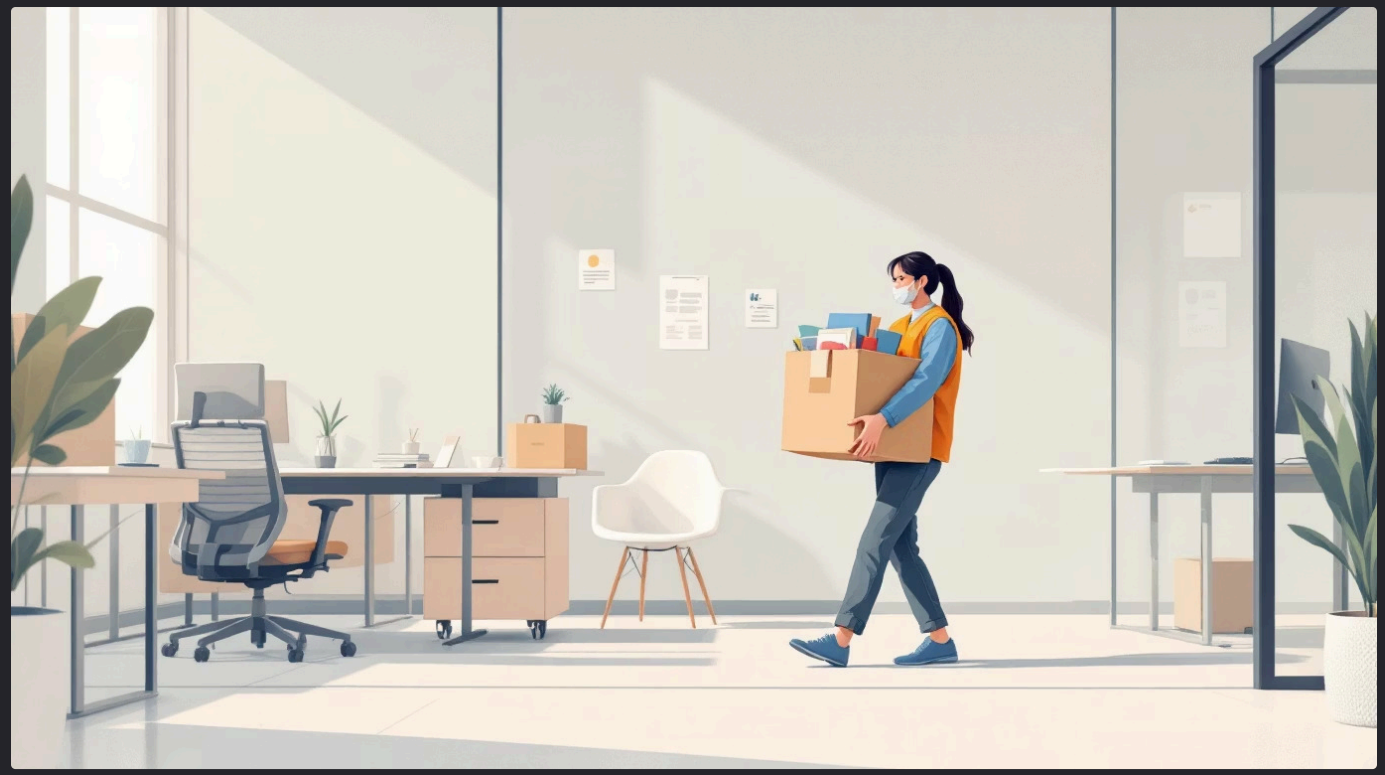

The Cost of Attrition

Every departure carries a hidden price — hiring, onboarding, and lost productivity.

The Challenge

Identify **key drivers** of employee attrition before they escalate

The Goal

Deliver **actionable insights** to reduce turnover and retain top talent

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Dataset & Tools

IBM HR Analytics

Industry-standard HR dataset

1,470 Employees

Comprehensive workforce sample

Power BI

Primary analytics & visualisation tool

DAX & Modelling

Data modelling, DAX measures, rich visualisations

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Workforce at a Glance

1,470	237	16.1%	$6.5K
Total Employees	Attrition Count	Attrition Rate	Avg Salary

7 yrs

Avg Tenure

- **R&D dominates** — 961 employees concentrated in research & development
- **Sales & Lab roles** form the largest workforce segments
- **Attrition is uneven** — concentrated in specific departments

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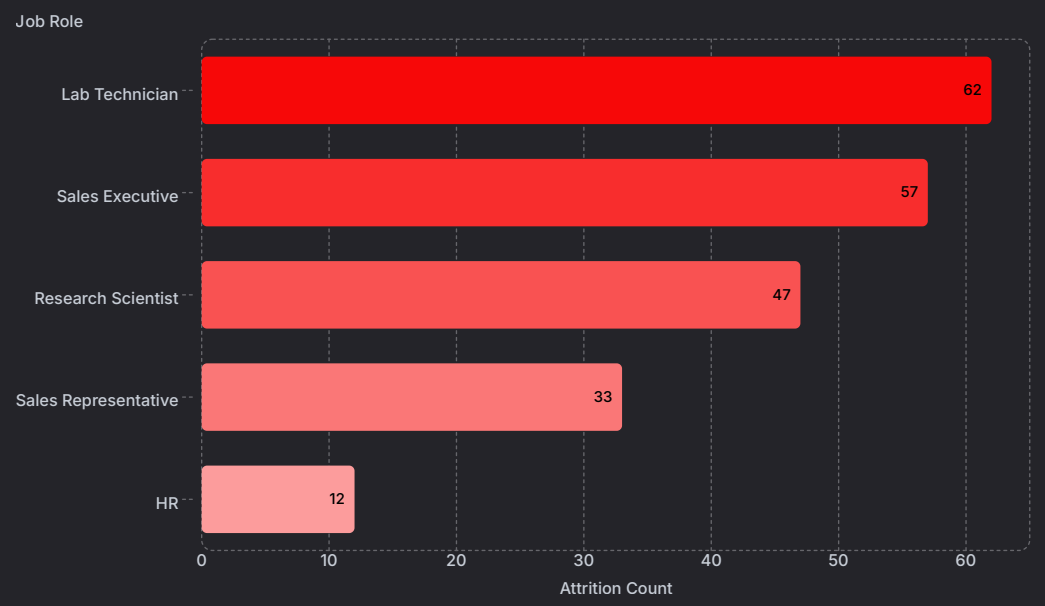
Attrition Analysis



Where Attrition Hits Hardest

Lab Technicians and Sales Executives account for nearly **half** of all attrition.

163 employees

Earning **up to $5K** — highest attrition salary band

Early Tenure

Attrition peaks in the **first few years** of employment

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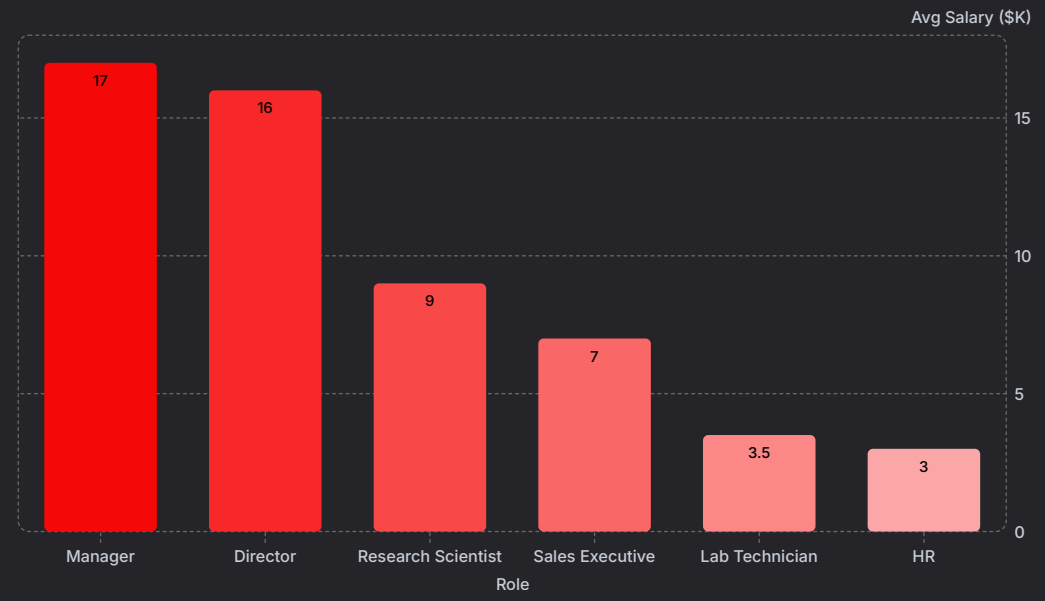
Compensation Landscape



Salary scales clearly with seniority — entry-level roles cluster near **$3K**, creating a retention risk.

Average

$6.5K — broad range across roles

Maximum

$20K — senior leadership tier

Median

$4,919 — majority earn below average

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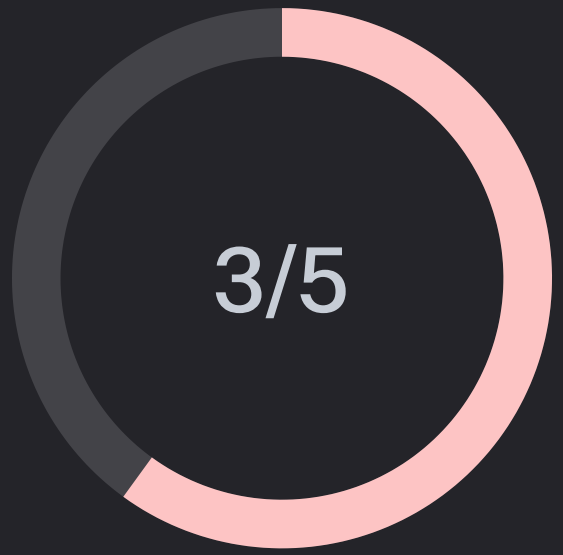
Performance & Satisfaction

Job Satisfaction

Below mid-scale — a significant concern

Performance Rating

~85% of employees rated at level 3



Work-Life Balance

893 employees rated level 3 — moderate

Most employees cluster around the mid-range across all three metrics — signalling systemic, not isolated, dissatisfaction.

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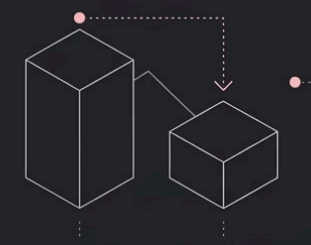
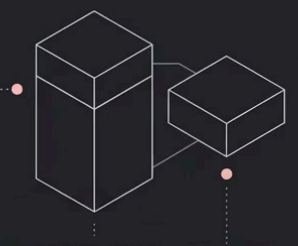
Key Drivers of Attrition

JOB SATISFACTION



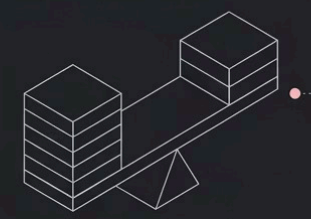
Low=30.3% High=13.9%

OVERTIME



Yes=33.1% No=10.8%

WORK-LIFE BALANCE



Poor balance=31%

JOB ROLE

Sales Representative=39.8%

The Numbers Tell the Story

- 3× higher attrition with overtime
- 39.8% attrition — Sales Representatives
- 30.3% attrition at lowest satisfaction

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Key Findings

Overtime = 3× Risk

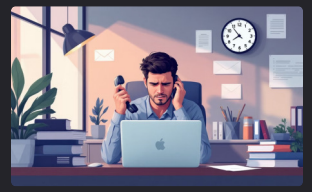
Overtime is the single strongest predictor of attrition

Low Satisfaction Drives Exits

Satisfaction level 1 triggers **30.3%** attrition rate

Entry Salary Band at Risk

Employees earning under **$5K** account for 163 departures



High-Risk Roles Identified

Sales Reps, Lab Technicians face disproportionate attrition

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Recommendations

Reduce Overtime

Cap excessive hours, implement workload redistribution policies

Boost Satisfaction

Launch engagement programmes targeting roles with low scores

Prioritise High-Risk Roles

Design tailored retention plans for Sales Reps & Lab Technicians

Improve Compensation

Review pay bands for sub-$5K earners — bridge the salary gap

Bottom Line: Data-driven HR analytics transforms attrition from a reactive cost into a proactive opportunity for retention and growth.

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